

Introduction

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Volker Schöffl in „El Dolphin“, Rodellar, Spain, Photo Kilian Reil

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Rock and sport climbing is currently evolving from an individual fringe sport into a popular mainstream sport. The worldwide boom of the popular subdiscipline of bouldering (ropeless climbing on low height boulders) is especially contributing to the overall increase in popularity. Indoor climbing centers are becoming more common and are no longer solely training facilities for specialists but have become a meeting place for leisure athletes, “weekend warriors,” and afterwork enthusiasts. The climbing gyms offer activities as children’s birthday parties, amateur competitions, and beginner classes for all age groups. Due to inclusion of climbing as a competition sport into the Tokyo Olympic program and the regular livestreaming of main climbing events such as World Cups, climbing now has a larger audience than ever. This upward trend has been constant over the last decade; there are more than 500.000 active climbers in Germany and more than two million in Europe [1]. Studies from the American Outdoor Industry Foundation report as many as nine million participants in rock climbing in the USA. The number of people who identify themselves mountaineers raise these numbers even higher, although the exact definition of a “mountaineer” can greatly influence the scope of the group [2].

In addition to the classic, world-renowned outdoor climbing areas such as Yosemite Valley (USA), the Alps (Europe), the Peak District (UK), and the Frankenjura (Germany), many new outdoor climbing areas have been discovered around the globe; rock climbing is now present on all continents and in almost every country. Many remote places such as Laos, the Seychelles, Oman, Reunion, or Antarctica have become well-known climbing destinations. While this globalization of outdoor climbing activities continues, more and more indoor climbing gyms are opening. Indoor climbing walls are present in almost every major city around the world, as well as in schools, rehabilitation centers, hospitals, army camps, and cruise liners. Consequently, indoor competition climbing has become popular not only on an international level but on local and regional levels as well. In the international competition scene, the climbing and training intensity

is constantly growing, and almost all top-level climbers are now full-time professional athletes. While this development is understandable in a Western society with a stable financial foundation, it is even more exciting to see the climbing boom spillover to developing countries. Climbing is a new sport in many countries such as India, China, Laos, or the Philippines. It is great to see that despite limited financial means, there is a strong and growing local climbing scene driving the sport forward in these places, especially since the sport was mostly inaugurated through traveling Western pioneering climbers like ourselves in Thailand, Laos, and other destinations in Southeast Asia in the early 1990s. We regularly receive emails and social media messages from injured climbers around the globe who have little access to sports medical facilities in developing countries. All of these climbers are connected by their love for climbing movements, thirst for adventure, and their desire to set and achieve new goals. These common interests provide a common ground for communication within the community between people of all ages from all over the world.

For sports medical doctors, climbing provides substantial professional challenges as its pathologies are very sport-specific and many of them are not present in other athletes. Although a common ankle sprain, a frequent injury in many other sports as well as climbing (especially bouldering), is treated by general sports medical and orthopedic trauma guidelines, other injuries such as pulley tears are climbing-specific and almost only seen in these athletes. Additionally, climbing has only been established as a professional sport for approximately 30 years, and there are still many unanswered questions about long-term developments and consequences. The increasing professionalism of Olympic competition climbing has led to a drastic increase of the overall training load and intensity [3, 4]. This will certainly lead to an increase in injury and overstrain incidence and could also affect long-term consequences [4, 5]. The change of the competition format to the “Olympic combined” mode for Tokyo 2020 (all athletes take part in all three competition disciplines: lead, boulder, and speed)

also influences the impact on the athletes; we found a certain change in injury epidemiology, e.g., an increase in hamstring or knee injuries and more frequent overstrain edema of the midcarpal bones [4–7]. On the other hand, the increased popularity of climbing as a leisure and recreational sport also leads to an increased occurrence of more severely graded trauma in beginners [8]. Based on the fact that many of these beginners are nonathletes, minor falls which would probably not injure a well-trained athlete may have serious consequences for these beginners. This may be due to a lack of core-stabilizing muscles, a lack of coordination, or other motoric deficits. The term “newbie syndrome” has been used to describe high-grade injuries in beginner climbers, for example, vertebral or tibial plateau fracture as a consequence of minor trauma (such as a fall from only 0.5 m height onto good protective mattresses) [8]. This also demonstrates that the demands on the sports medical teams increase and adapt with the changing direction of the climbing sport itself.

The aim of this book is to demonstrate the climbing-specific anatomical and biomechanical specifics and to analyze the specific injuries and overstrains as well as the medical, gynecological, and pediatric aspects of the sport. While our book “One Move Too Many” (Schöffl V., Hochholzer T., Lightner S. Jr.) is aimed at trainers, athletes, and laypeople, this book is based on the German “Klettermedizin” (Springer) and targets medical professionals and scientists with its main focus on evidence-based knowledge in climbing medicine.

1.1 History

The various versions of modern sport climbing have all developed from mountaineering, a sport which was established in the European Alps. By the mid-1980s, the popularity of climbing had spread globally and had diversified to include new categories, such as ice climbing, bouldering, speed, and aid climbing. In many instances, the style in which a route is climbed and the difficulty of the route itself became more important

than reaching the summit itself [9]. Style refers to either free climbing, which is defined as a climb completed using only the rock face formations, or aid climbing, in which various devices such as a sling ladder can be anchored into the wall in order to assist the ascent [10]. Simultaneously to the developments in sport climbing, the mountaineering routes became increasingly challenging, both mentally and physically, and started to fall into the definition of extreme climbing.

The basic idea of “free climbing” probably originated from the Elbsandstein region in the Eastern part of Germany, where the modern style-oriented free climbing approach has been used for over 100 years. In the Elbsandstein region, the use of pitons and bolts for upward ascent was considered cheating, so only the athlete’s hands and feet could be used to scale a face. In 1918, Emanuel Strubich free climbed the “West Arete” at the Wilde Kopf, establishing the first free climb to be graded UIAA 7–, or 5.10 on the Yosemite decimal scale (YDS). It was a monumental achievement upon which other free climbers like Berndt Arnold would build in the future. However, with the beginning of the Cold War, the East German climbing scene was lost to the rest of the world, and the concept of trying to “free” a climb disappeared with it [2].

In other climbing areas of the world, the sports of rock climbing and mountaineering went through a radical transformation in the late 1960s and early 1970s. The stone master generation of climbers in Yosemite Valley began looking at the route rather than the summit as the most important part of the climb. With that, the style of climbing was brought into question, and the quest for ascending a route specifically by “free climbing” was born. The rope, pitons, and new pieces of clean-climbing protection called nuts (and later friends) were soon being used to arrest a fall rather than to aid the ascent. The grades of difficulty in free climbing consequently began to jump off the scale [2].

Two Americans, both climbing in relative obscurity to the well-known scene in Yosemite, made some of the biggest steps in free climbing. John Gill brought gymnastic skills and many gymnastics training techniques to the sport in the

early 1960s. Using the previously unseen gymnasts' chalk to keep his hands free of sweat, Gill established boulder problems that would not be repeated for a decade or more. One of these problems, "The Thimble," would be a four or five bolt sport route today and was undoubtedly the first UIAA 8, or 5.11+ YDS [2]. Tony Yaniro followed in his footsteps, approaching the sport with training techniques he had learned from the US Olympic Team. Yaniro approached training systematically and used specially designed equipment, including an early version of the campus board, to improve his skills as a free climber. All of this training paid off in 1979 with the first ascent of "The Grand Illusion," the world's first UIAA 10- (5.13c YDS) [2]. In Central Europe, Reinhard Karl and Helmut Kiene opened the previously limited UIAA scale with their legendary first ascent of the "Pumprisse" (Fleischbankpfeiler, Wilder Kaiser). The seventh UIAA grade had finally also been climbed in an alpine environment. The British climber Jerry Moffat added the route "The Face" (Germany) to the tenth grade in 1983 before the German climber Wolfgang Güllich began to push the grades even higher. He established routes such as "Wallstreet" (UIAA 11-, 5.14b YDS) in 1987 and "Action Direct" (UIAA 11, 5.14d YDS) in 1993, a route that didn't see a second ascent for years [2]. Most recently, Chris Sharma, Adam Ondra, and Alex Megos have further raised the bar with routes that are solidly graded in the UIAA 12 (YDS 5.15) range. Adam Ondra's route "Silence," Flatanger, Norway, established in 2017, is currently considered the hardest rock climb in the world (UIAA 12, YDS 5.15d).

1.2 Terminology

The vocabulary of rock and sport climbing uses many different terms both for its subdisciplines and for the various ways a in which a climb is ascended. Rock climbing is a multidiscipline sport and as such, different levels of risk must be considered. The risks depend on the subdiscipline examined, the climber's experience and skills, the grade of a route's difficulty, equipment,

climbing surface (type of rock or ice, artificial indoor wall, scree), remoteness of location, altitude, and weather [11]. It is common for climbers to regularly participate in more than one climbing subdiscipline, raising the overall participation time in the sport and subsequently increasing the risk of injury. In the following paragraphs, the various subdisciplines will be discussed in further detail.

Sport climbing or free climbing (Fig. 1.1) is similar to gymnastics in many ways. It requires a combination of flexibility, finger strength, overall upper body strength, and endurance in a manner dictated by the route climbed. The climbing is slightly prescriptive as the climber ascends toward mostly permanently fixed **anchors**, such as predrilled **bolts**, to clip their rope into for **protection**. The route length can range from 10 m over 100 m with fixed anchors usually placed 2–5 m apart. Falls are frequent, are practiced, and are mostly harmless and considered a common part of the discipline (Fig. 1.2). Physical hazards (rock fall, weather changes, etc.) are minimal, and neglecting to wear a climbing helmet is widely accepted. The term free climbing is often misused and misunderstood; it reflects the fact that the climber does not use any artificial aid but climbs a route from ground to top without loading the belay equipment (rope, quickdraws, etc.). The climber does not pull or rest on gear and bolts, not climbing unprotected but free from technical aid. Climbing a route without any protection is referred to as "free solo climbing," whereas "solo climbing" is defined as a single



Fig. 1.1 Sport climbing. Photo Kilian Reil



Fig. 1.2 Normal falls into the rope are frequent and trained for in modern sport climbing. Photo Kilian Reil



Fig. 1.3 Bouldering. Photo Enrico Haase

person completing the climb, protected by rope and gear in a very specialized technique. The opposite of free climbing is aid climbing, in which hooks, pitons, bolts, and other kinds of gear are used as active aid to ascend a route. A sport climb (free climb) is only counted as completed if it is performed as a so-called redpoint, defined as having climbed the route from the bottom to the top and without loading the belay devices. If a climber climbs a route on the first try without any prior knowledge of the route, it is called “on sight”; if the route is climbed on the first try with information on the route (e.g., seen someone climb it, analyzed video footage, etc.), it is called a “flash” ascent. Climbing a route with rope protection from above is called “top rope.”

Bouldering is defined as ropeless climbing involving a usually short and low sequence of powerful and technical moves to complete the graded route on large boulders at a height from which the climber can safely jump down. However, these boulders can be up to 10 m high, classifying these as a “**highball boulder**.” Bouldering (Fig. 1.3) can be performed without a partner and with minimal equipment—climbing shoes and a crash pad (a portable mattress for fall protection). Falling onto one’s feet or body is common in bouldering, whether a route is completed (defined as “topped”) or not.

Traditional (alpine) climbing emphasizes the skills necessary for establishing routes in an exploratory fashion outdoors, performed normally in a group of two climbers. The lead climber typically ascends a section of rock while



Fig. 1.4 Traditional climbing. Alex Luger in Moab, USA. Photo Archive Adidas

placing removable protective gear where possible along the climb (Fig. 1.4). Therefore, falls can be much longer than those experienced when sport climbing. Unreliable fixed pitons may occasionally be found on older established routes. As physical hazards are likely, the use of a helmet is considered mandatory (Fig. 1.5). Above altitudes of around 2500 m, physiological altitude-induced adaptations must also be factored into the climbs [11].

Indoor climbing is performed on artificial structures that try to mimic outdoor climbing in a more controlled environment. As physical hazards are almost totally eliminated, indoor climbing has become an extracurricular sport in many countries. National and international competitions are held on indoor walls and involve three major disciplines—lead climbing (i.e., sport climbing), speed, and bouldering (Fig. 1.6).



Fig. 1.5 Alpine climbing. Mayan Smith-Gobat, El Capitan, Yosemite Valley, USA. Photo Archive Adidas



Fig. 1.6 Indoor bouldering. Photo Kilian Reil

Bouldering is performed above thick foam mat flooring, speed climbing in top rope, and lead climbing in an on-sight style [11].

1.3 Grading

The most common of the many various grading scales are the UIAA, the French, and the Yosemite decimal scale. In bouldering, different grading scales exist; the French Fontainebleau scale (Fb scale), and the US Vermin scale (Hueco scale or V-scale) are the most important. For scientific analysis, the decimal UIAA scale [12] or the International Rock Climbing Research Associations scale [13] is recommended.

1.4 Equipment

In the early times of climbing, classic heavy mountaineering boots were worn for climbing in alpine regions as well as on rock faces. The first real climbing shoe (the EB) with a friction sole only entered the market at the beginning of the 1980s. A common characteristic of all climbing shoes is that they have to fit very tightly in order to obtain optimal contact to the rock, which often leads to health problems, such as callosity, toenail infections, or even long-term conditions such as hallux valgus deformities [14]. The introduction of bolts was an important factor for the explosive development of the sport climbing grades; falls into the rope are now common for sport climbers [15]. The structure of climbing harnesses also has changed dramatically, from the combination of a chest and seat harness traditionally used in mountaineering to the pure seat harness currently used sport climbing [15, 16]. This allows injury-free falls with the maximum freedom of movement while climbing. Modern belay devices are semiautomatic, targeting the reduction of human error while belaying. Nevertheless, the correct techniques of lead and boulder falling, spotting, and belaying are essential. The bolts, ropes, harnesses, and other equipment used for climbing should be approved by the UIAA safeties commission [17]. In bouldering, crash pads, portable foldable mattresses with various foam layers, are placed underneath the climber to protect the falls (Fig. 1.7). “Spotting” is also an important component injury prevention, in which other climbers protect the falling climber by directing the climber’s fall toward the mats in order to minimize the severity of the impact. There have also been reports of injuries to the “spotter” caused by the falling climber (Fig. 1.8).



Fig. 1.7 Crash pads for injury prevention in bouldering. Photo Enrico Haase



Fig. 1.8 Spotting in bouldering. Photo Kilian Reil

1.5 Competition Climbing

Since the 1970s, speed climbing competitions have been held regularly in East European regions. In the mid-1980s, climbing competitions became more frequent, with the first ever German Climbing Championship held in 1985 in a beer hall in Munich (“Salvator Keller”). The first, still unofficial World Championship took place in 1991 in Frankfurt. Since then, World Cups, national and international, championships have taken place regularly. The World Championships in Munich in 2005 already hosted more than 500 participants from 55 countries. The international body of competition climbing is the International Federation of Sport Climbing, founded in 2009 in Paris. Parallel to these official IFSC competitions, various large events, so-called masters are also held, receiving substantial media coverage, such as the “Adidas Rockstars” event in Stuttgart, Germany, or the “Rockmasters” in Arco, Italy.

All competitive events are performed on artificial walls, both indoors and outdoors. The increased presence of climbing as a competition sport peaks with the inclusion into the Olympic program of Tokyo 2020 and Paris 2024.

1.6 Disciplines

Three disciplines are currently being held in national and international competition formats: **lead** (climbing with rope protection), **speed** (maximum speed climbing on a standardized route in one-on-one mode), and **bouldering** (climbing at lower heights with mattress floor protection). In addition, a triathlon of all three disciplines (**Olympic Combined**) has been launched recently, representing the Olympic competition format. In addition to the official disciplines, various other events such as marathon climbing or boulder nights have become increasingly popular, hosted mostly in local and regional competitions with a focus on entertainment and recreational sports. Some international master competitions are also offering new disciplines such as “duel climbing” or “after work” formats. In “duel climbing,” both athletes compete on identical routes, either in boulder or lead, similar to the classic speed climbing competitions. In the “after work” format, climbers have the chance to practice the route for a certain time frame prior to the competition and have one “redpoint” attempt for the final competition. The three standard competition disciplines will be further explained below.

1.7 Lead Climbing (On-Sight)

In lead climbing, the athletes rope climb a route on-sight as far as they can, ideally all the way to the top of the artificial competition wall. International competition routes are a minimum of 15 m in length and are set specifically for the event by certified route setters. Prior to climbing, all athletes participate in a collective route inspection lasting 6 min. The athletes are then placed in an isolation zone until their turn to climb, guaranteeing that they cannot see how the other



Fig. 1.9 German Championship Lead Climbing (DAV, German Alpine Club). Photo Marco Kost



Fig. 1.10 Competition bouldering at “Adidas Rockstars.” Photo C. Waldegger

climbers attempt the route and removing an unfair advantage for the later climbers. All climbers have the same time limit of 6–8 min on the route. The first criterion for the result is the overall climbing length, measured to the last handhold the climber held. If there is a tie based on progress up the route, the time taken to reach the hold determines the ranking (Fig. 1.9).

1.8 Bouldering

Bouldering is performed without a rope above thick foam mats, usually at a height from which the climber can safely jump down. The boulder problems are also set specifically for the competition and are between 3 and 4 m high. The results are calculated based on the number of completed boulders, number of bonus holds reached (earlier

called “zone”), and the number of attempts to complete the boulders as well as the number of attempts to reach the bonus holds. A bonus hold is usually a hold halfway through the boulder problem and is specially marked [18] (Fig. 1.10).

1.9 Speed

The overall goal of speed climbing is to top-rope climb to the top of the route and to hit the buzzer before the opponent. Per round, two climbers compete against each another on identical adjacent routes in a knockout system. Both climbers climb both sides once with a short break in between. The times from both ascents are summed and the faster climber reaches the next round. In contradiction to lead and boulder, the speed routes have remained exactly the same



Fig. 1.11 German Championship Speed Climbing (DAV, German Alpine Club). Photo T. Schermer

since the 2005 World Championships, with standardized wall angulation, normed handholds, and either 10- or 15-m length. This standardization allows for the times to be compared over all competitions and for world records to be set [18] (Fig. 1.11).

In addition to these standard disciplines, climbing is constantly developing and new disciplines are evolving. “Deep-water soloing” or “psicobloc” is a new subdiscipline in which routes are free solo climbed above water, usually oceans. These attempts often end in spectacular falls into the water below. There are also psicobloc competitions held on artificial walls above public pools. Additionally, solo and speed big-wall ascents are evolving and drawing attention from the media. As climbing doctors, we need to adapt to the dynamic evolution within our sport, which consequently presents us with new pathologies and injuries. This is the fascinating part!

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